

OLYMPICS

Data Analysis

Under the guidance of

Dr. Dootika Vats

GROUP 22

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Motivation

- Olympics is the most prestigious sports event and it's a pride for the athletes to participate in this event and win medals for their nation.
- Loads of data was available online on Olympics, so we could do a lot of things and play around to figure out the various factors that could be responsible for winning more medals and make some interesting visualizations.



Type of data collected

- To analyze the Olympic medal data, we collected the count of medals won by different countries over the years.
- We also collected the data for various other factors like, Literacy rate of the country, Population, Life Expectancy, BMI, Per capita alcohol consumption, etc.
- The data containing the list of athletes and their sex, age, height, weight, medals won was collected.

We have primarily three datasets:

- Total number of medals (gold, silver and bronze) till date and some other factors. (101 x 15)
- Year-wise medals for each countries, and the host nation. (2071 x 8)
- The age, sex, height, weight of the athletes who have participated in the events over the years. (206156 x 5)

DATA EXTRACTION

- Data was extracted from various websites using the R library "Rvest" .
- Some data were directly downloaded from websites in .csv format.

THE FOLLOWING SOURCES WERE USED

- https://en.wikipedia.org/wiki/All-time_Olympic_Games_medal_table
 - <https://worldpopulationreview.com/country-rankings/olympic-medals-by-country>
 - https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results?select=athlete_events.csv
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Libraries Used :

- ggplot2
 - dplyr
 - rvest
 - tidyverse
 - gganimate
 - shiny
 - shinydashboard
 - shinydashboardPlus
 - hrbrthemes
 - ggthemes
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Glimpse of the data

Nation	Gold	Silver	Bronze	Total	Literacy	pop
Algeria	5	4	8	17	79.6084	44903.225
Argentina	21	26	30	77	98.0900	45510.318
Azerbaijan	7	14	28	49	99.8053	10358.074
Azerbaijan	7	14	28	49	99.8053	10358.074
Barbados	0	0	1	1	99.7000	281.635

year	Rank	Nation	Gold	Silver	Bronze	Total
1896	1	United States	11	7	2	20
1896	2	Greece	10	18	19	47
1896	3	Germany	6	5	2	13
1896	4	France	5	4	2	11
1896	5	Great Britain	2	3	2	7

Sex	Age	Height	Weight	Nation
M	24	180	80.0	China
M	23	170	60.0	China
F	21	185	82.0	Netherlands
F	21	185	82.0	Netherlands
F	25	185	82.0	Netherlands
F	25	185	82.0	Netherlands
F	27	185	82.0	Netherlands
F	27	185	82.0	Netherlands
M	31	188	75.0	United States
M	31	188	75.0	United States
M	31	188	75.0	United States

Challenges Faced

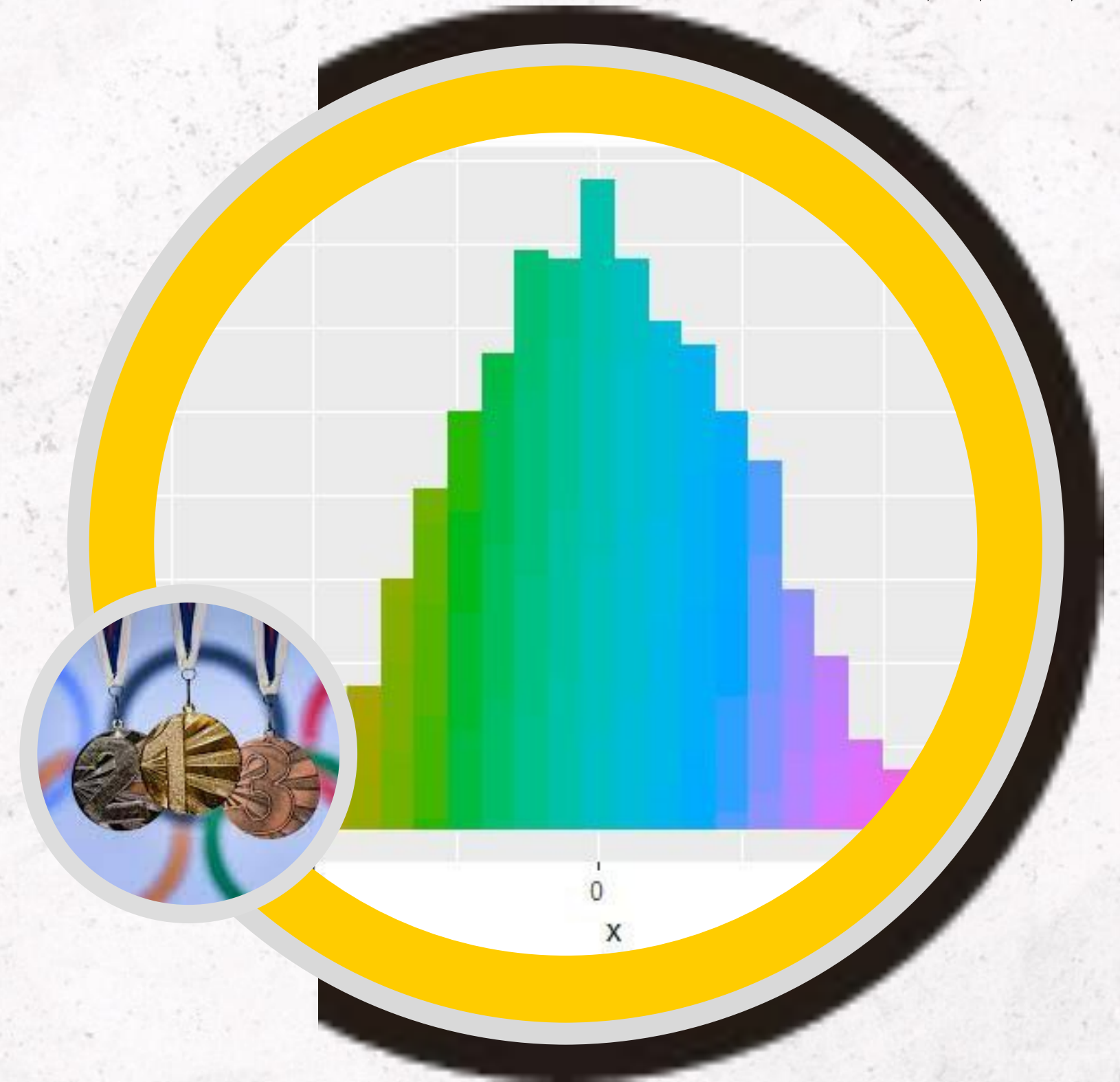
- Obtaining the data was easy but they had many irregularities. There were rows of the same country carrying same data but had different name for the country.
 - The names of the country's took some time to clean.
 - Since we used many factors to find correlation, data for all factors was not easily available for each nation. So we had to manually find some data and fill in the missing values.
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Key questions of interest

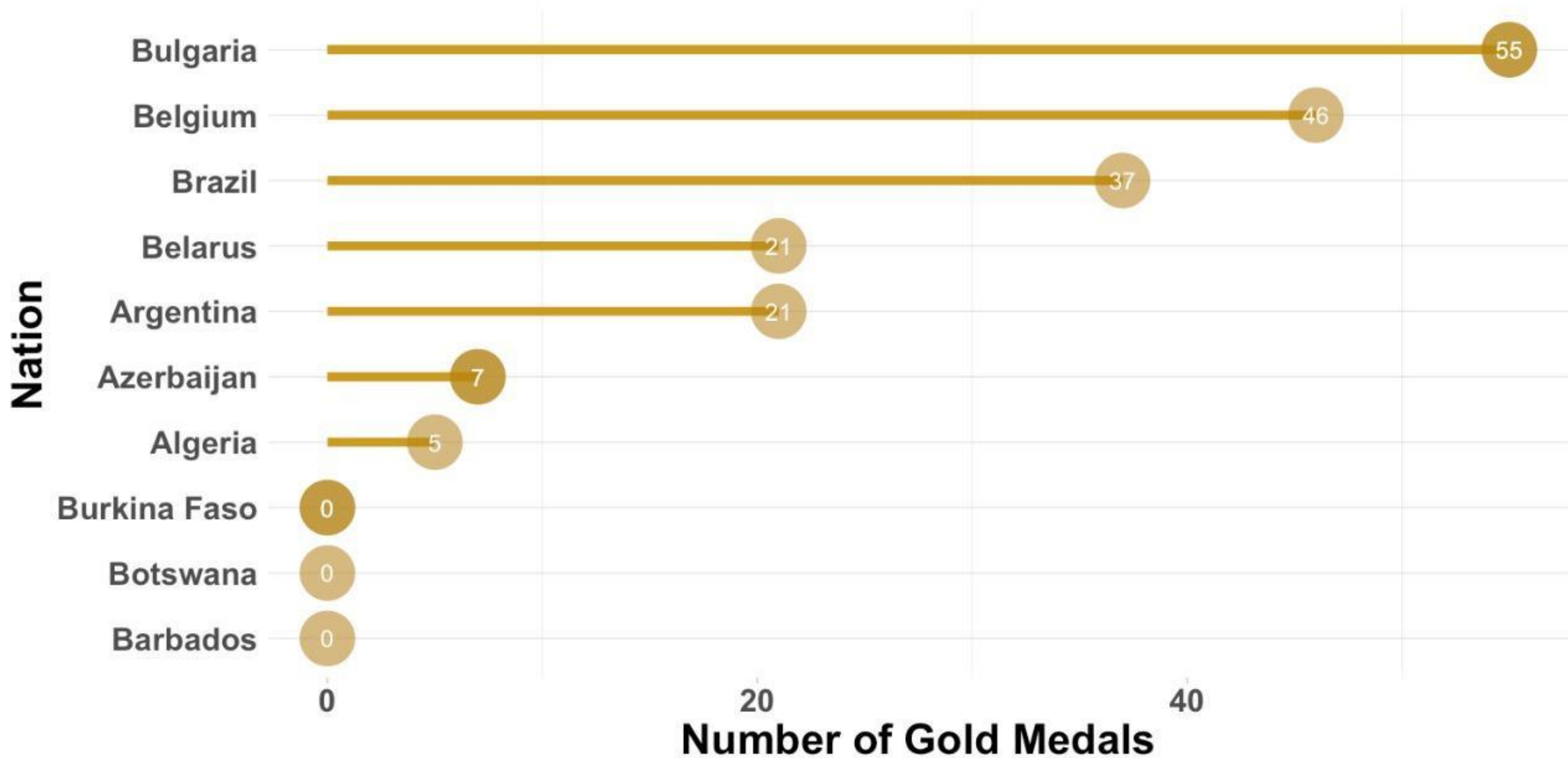
- Which are the top performing countries in Olympics and does the various factors like life expectancy, literacy rate, year of independence, etc. affect the medal count of a country ?
- Has popularity of Olympics increased over the years ?
- What is the age, height and weight distribution of the athletes ?
- Does hosting the event increase the chances of winning more medals ?

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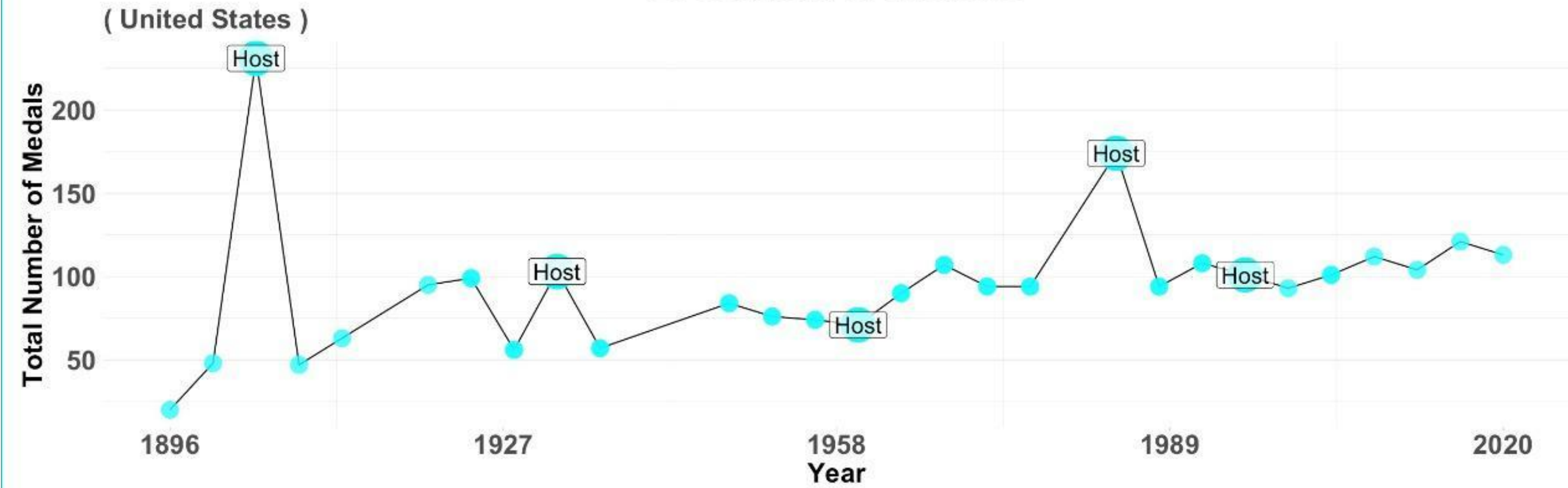
Some insightful plots



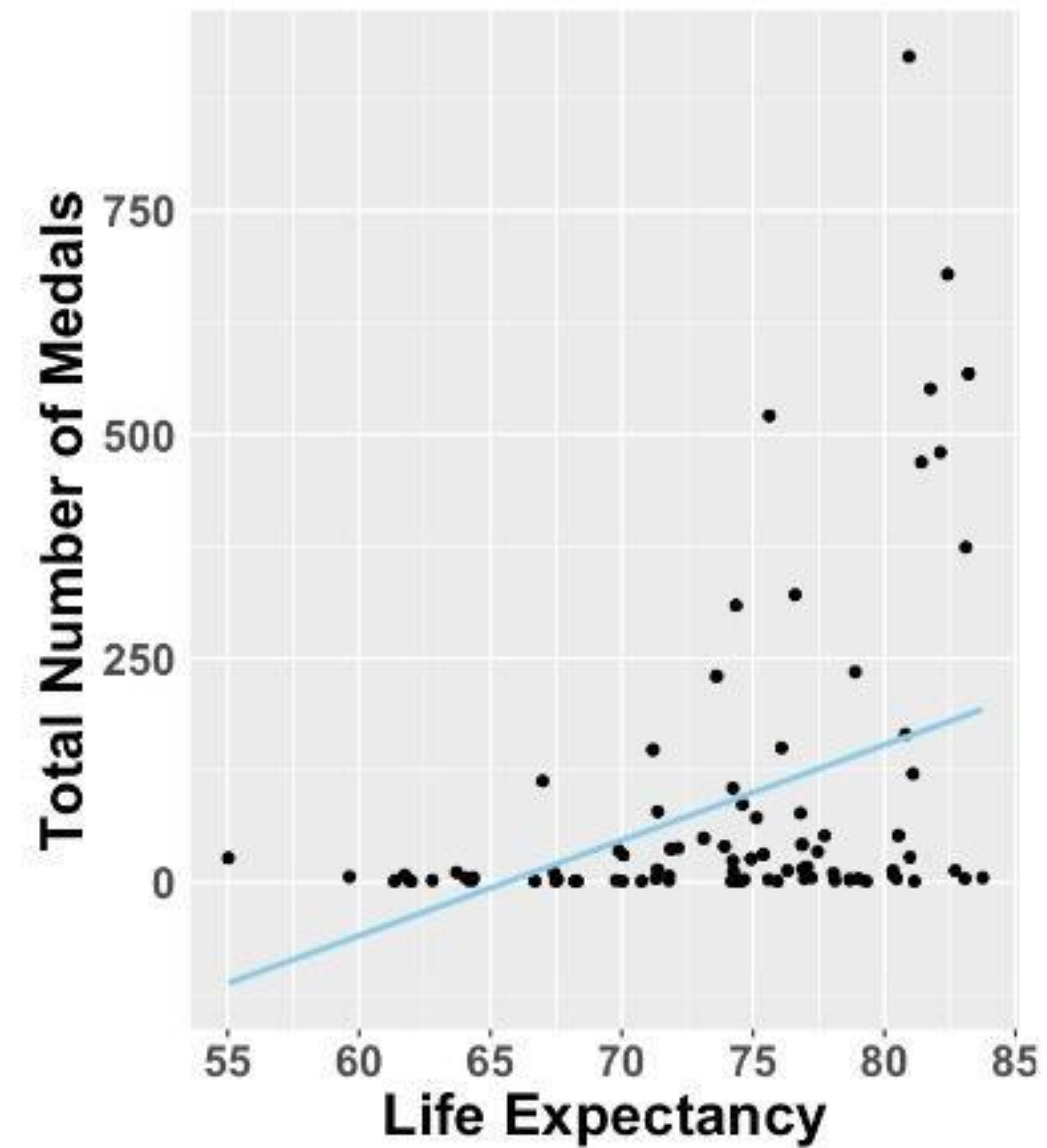
Race of Nations for GOLD



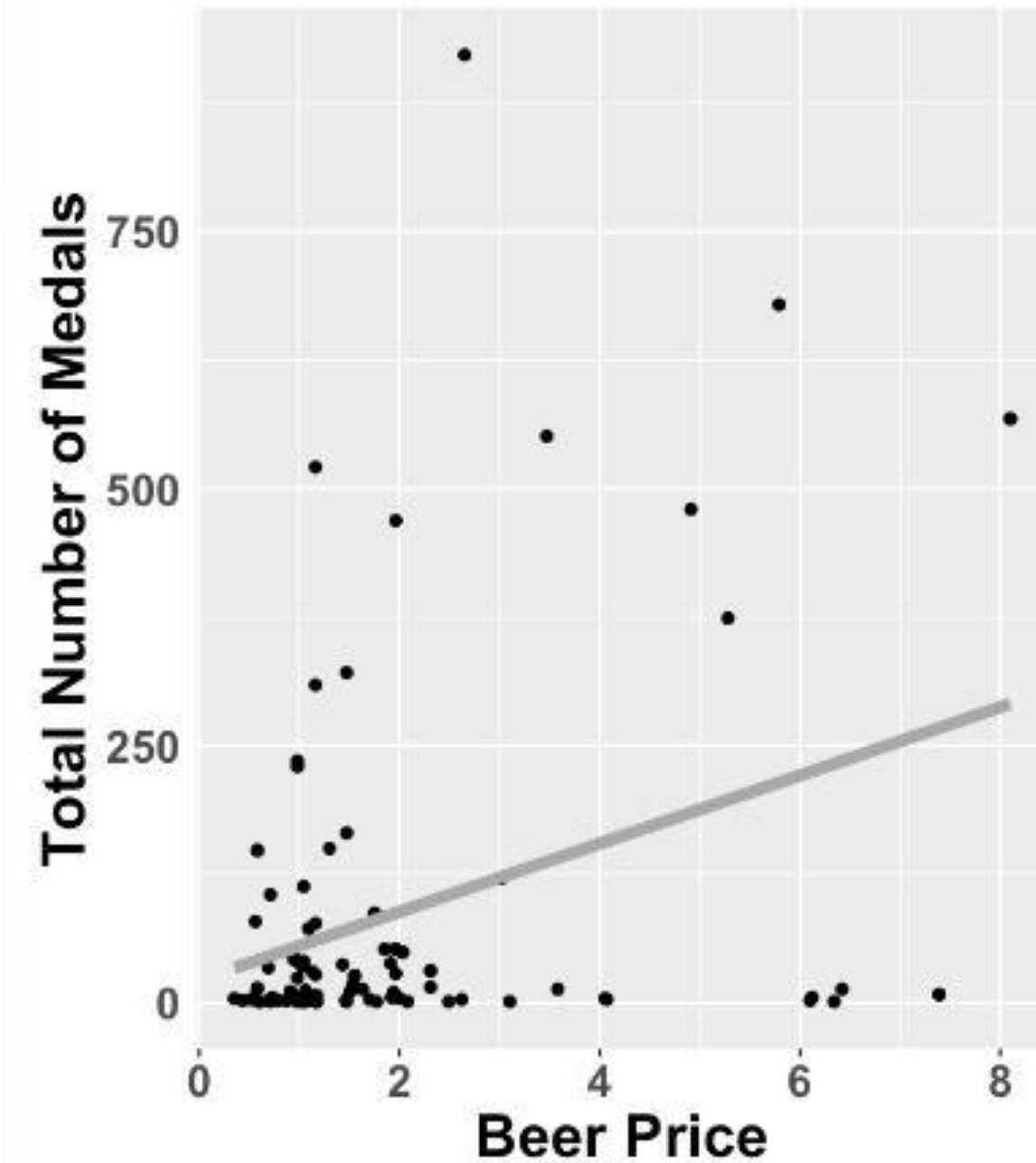
Evolution of Medals



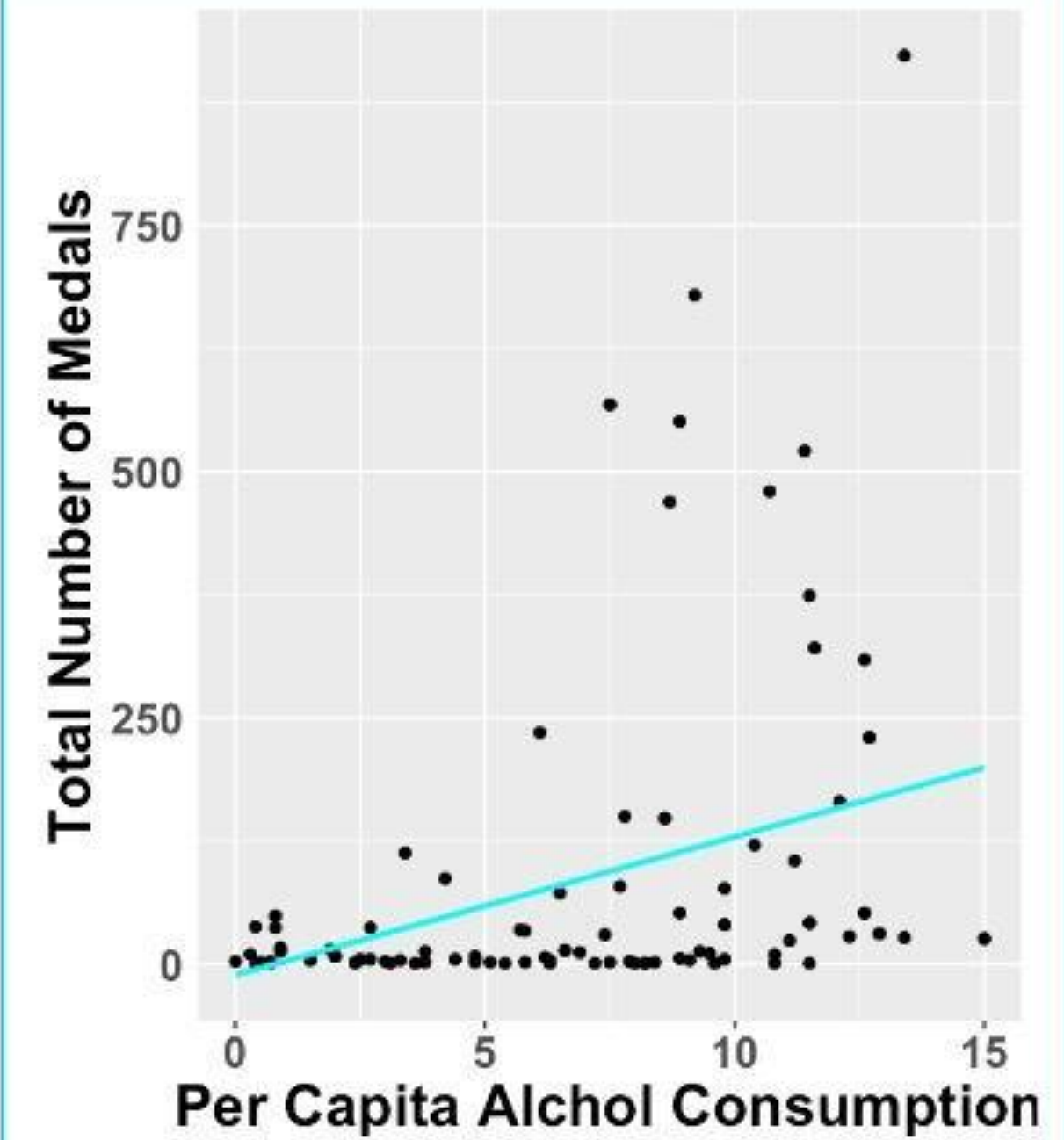
Life Expectancy (Cor = 0.3983424)



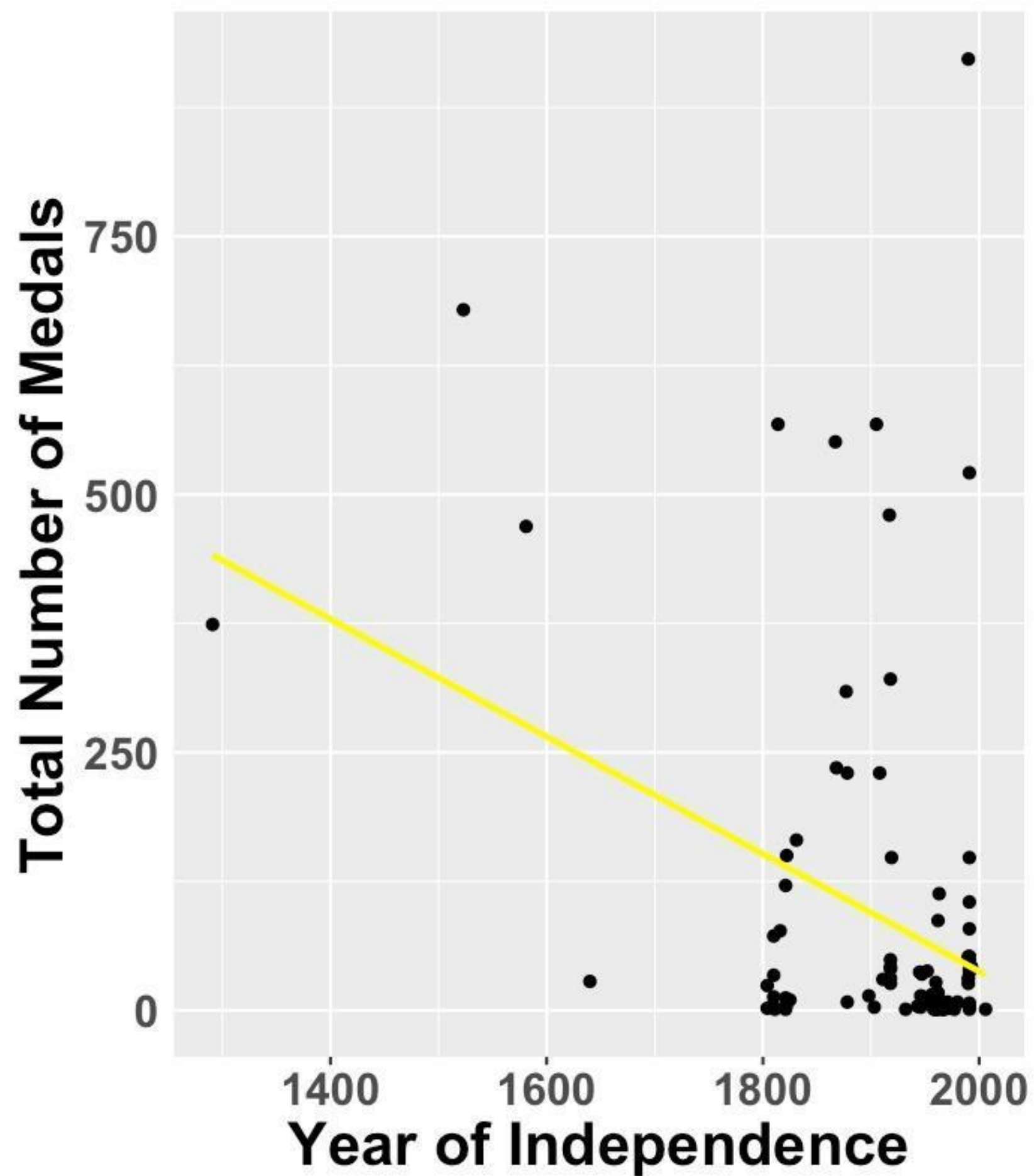
Beer Price (in USD) (Cor = 0.3507118)



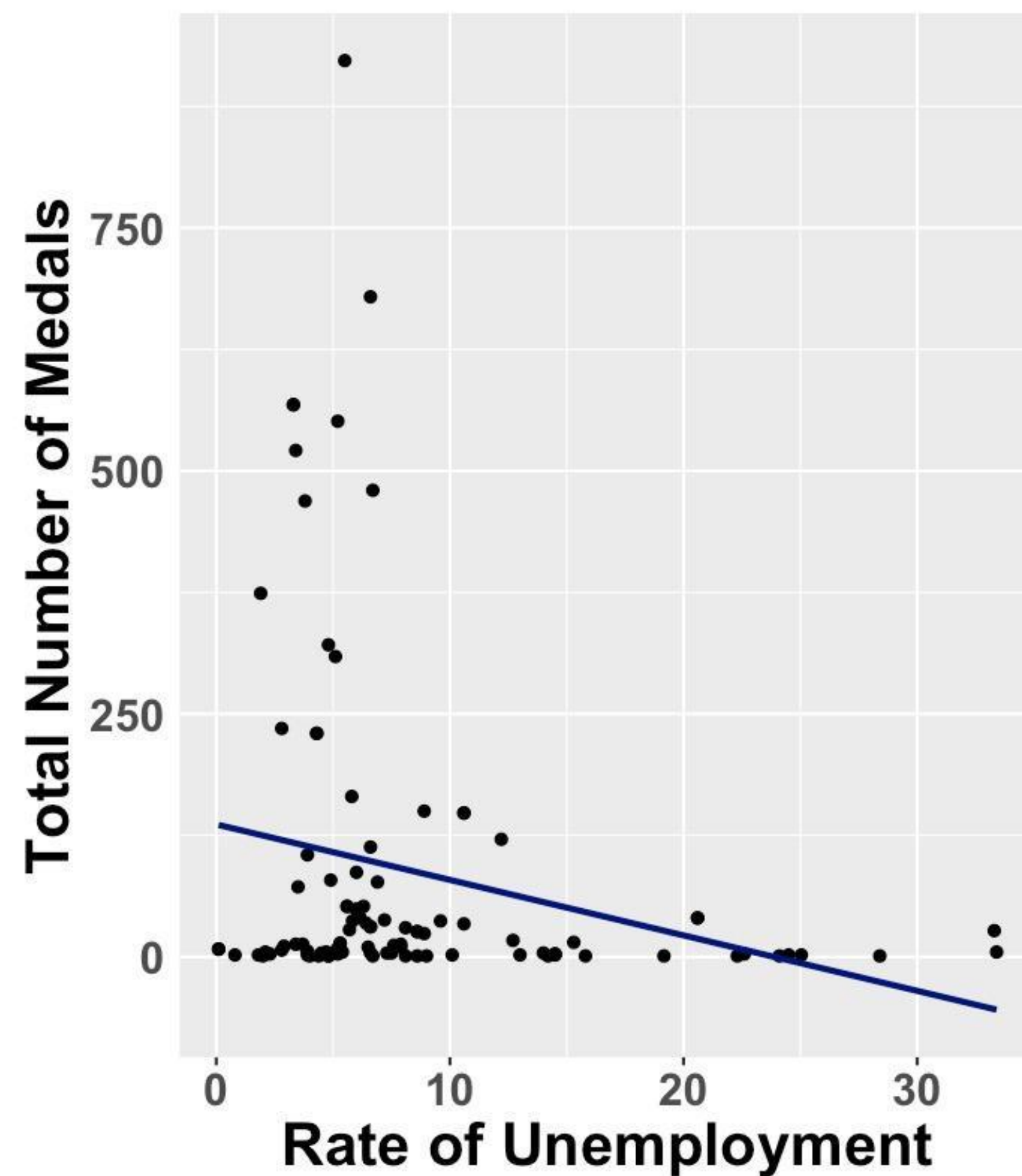
Per Capita Alcohol Consumption (Cor = 0.3381)



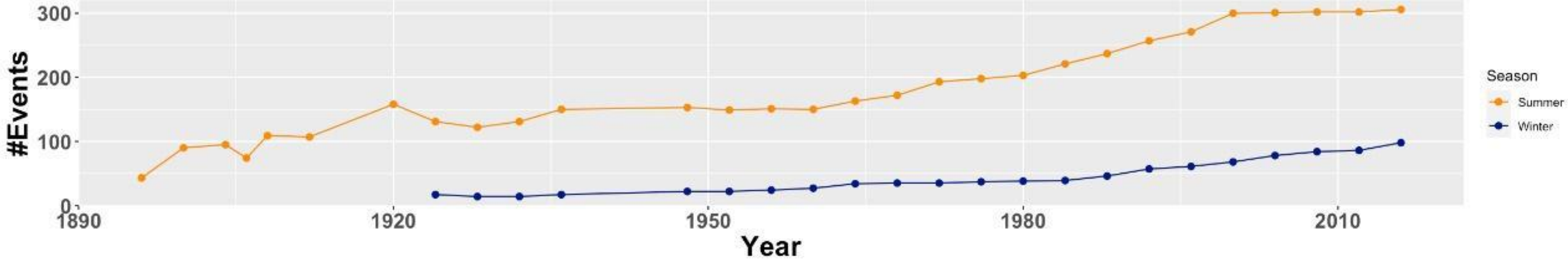
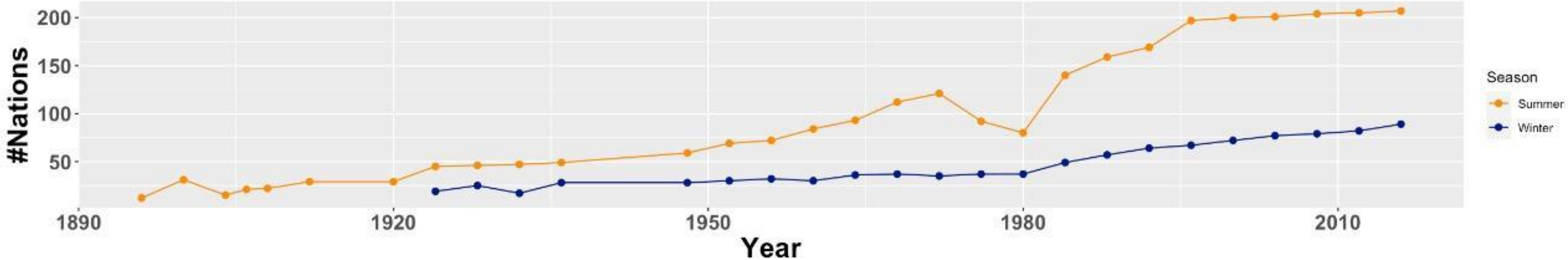
Year of Independence (Cor = -0.3535347 –



Rate of Unemployment (Cor = -0.2284803)



Total number of Nations and Events over time.



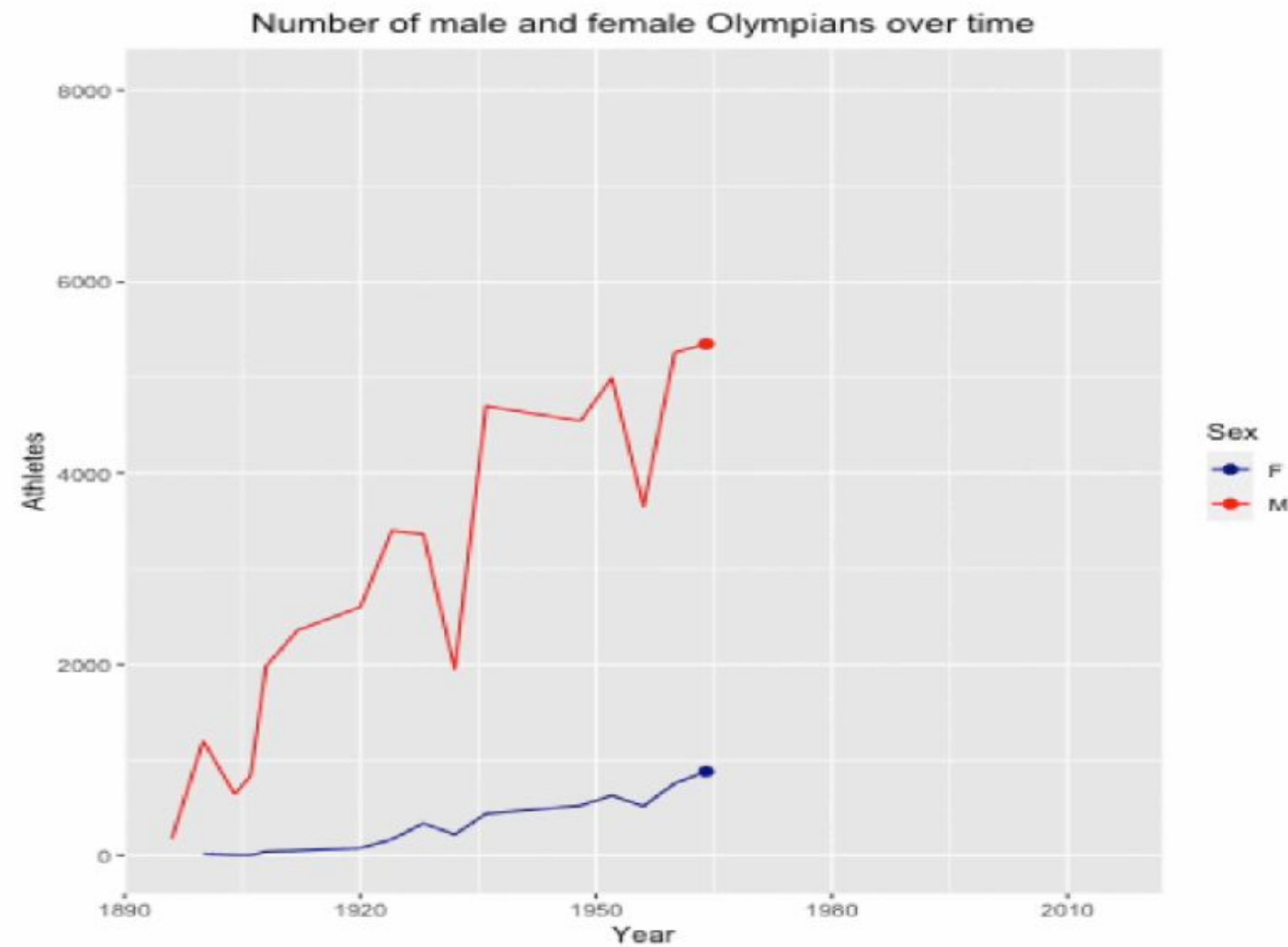


SHINY APP

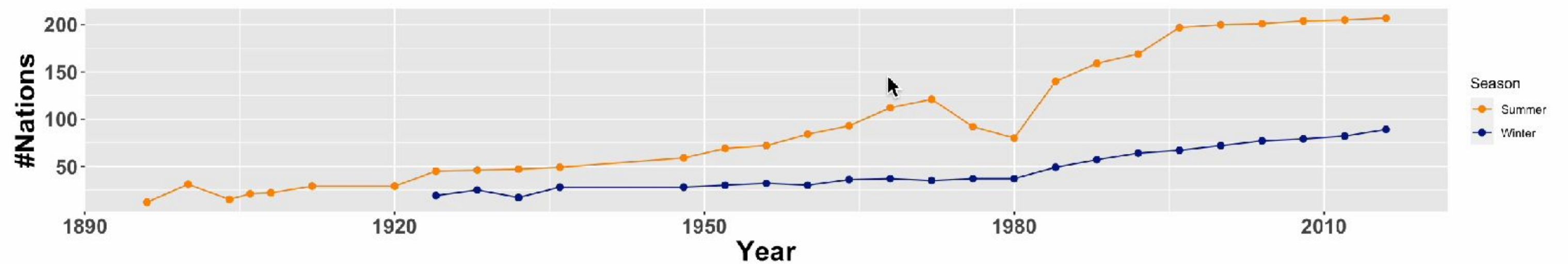


[Introduction](#)[Race for Medals](#)[Evolution of Medals](#)[Olympics Data](#)[Distribution](#)[Interesting Factors](#)[Source Code](#)

Athletes count



Total number of Nations and Events over time.



CONCLUSIONS:

- It can be clearly seen from the plots that there has been an exponential increase in the participation of countries and athletes. Also the number of events has increased, which shows that the popularity of the Olympics has increased over the years.
 - It can also be concluded that the countries which got independence early tend to win more medals.
 - The countries having higher life expectancy, higher per capita alcohol consumption won more medals than others.
 - And the medal count decreased with increase in unemployment rate
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CONCLUSIONS:

- Germany and Sweden were among the top performing countries.
 - With few nations we can observe that when they hosted the Olympics their Medals won sought to increase. (USA, Sweden, etc)
 - The Average age of Olympians was in the range of 24-28.(Highest was 72)
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We would like to acknowledge

Dr. Dootika Vats

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And special thanks to all the TA's

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THE TEAM

